Zihan Zhou



AI for Chemistry / Molecular Generation / AI for Science Research
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Ph.D. student in Data Science focused on generative models for scientific applications, including single-step retrosynthesis, protein sequence generation, and modeling with physical priors. Publications in TPAMI, ICML, NeurIPS, and ICLR.

Education

The Chinese University of Hong Kong, Shenzhen | Ph.D. in Data Science Sep. 2023 - Jun. 2027 (expected)

Advisor: Prof. Tianshu Yu

The Chinese University of Hong Kong, Shenzhen | B.Sc. Mathematics and Applied Mathematics Sep. 2019 - Jun. 2023

GPA: 3.513 / 4.0 | First Class Honors

Research Internships

ByteDance Seed | AI for Science Intern Aug. 2026 - Present

Conduct research in AI for Science.

Tencent 腾讯 Tencent Life Sciences Lab | Applied Research Intern Jun. 2026 - Aug. 2026

Generative modeling: Implemented algorithms for controllable protein sequence generation and editing; contributed to conditional modeling and the development and refinement of training and sampling pipelines.

Evaluation: Assessed candidate sequences and generation methods by target condition satisfaction, extent of sequence edits, and predicted structural quality.

Shanghai AI Laboratory | AI for Science Research Intern (Retrosynthesis) Aug. 2025 - May 2026

Single-step retrosynthesis: Led development and maintenance of the training, evaluation, and experiment codebase; built reproducible workflows for comparative studies and ablations.

Shanghai AI Laboratory | AI for Science Research Intern (Proteins) Jun. 2025 - Dec. 2025

Protein generation: Contributed to architecture design and large-scale training of MSA-conditioned generative models; compared conditioning and sampling methods and evaluated predicted structural quality.

Selected Research Projects

Reaction-Center-Guided Molecular Retrosynthesis ICML 2026

Shanghai AI Laboratory | Aug. 2025 - May 2026

- **Problem and approach:** Predicted reactants from products for single-step retrosynthesis. Prioritized reaction-center atoms to introduce a structural inductive bias, combining a graph diffusion Transformer with discrete flow matching.
- **Contributions:** Built the training and evaluation codebase from scratch and led its development and maintenance. Designed a Hydra-based experiment framework for reproducible ablations on atom ordering, model size, and sampling steps.
- **Results:** Achieved Top-1 accuracy of **61.2% / 51.3%** on USPTO-50k / USPTO-Full. Compared with prior diffusion methods evaluated in the paper, training was **6 times faster** with **1/25** as many sampling steps. With appropriate atom ordering, a **280K**-parameter model matched a **65M**-parameter baseline without this bias.

Protein Sequence Generation: MSA Modeling and Property-Controlled Editing

MSA modeling: Shanghai AI Laboratory, Jun. - Dec. 2025; controllable editing: Tencent, Jun. - Aug. 2026

- **MSA-conditioned modeling:** Adapted the ESM-2 Transformer into a DiT and compared BFN and DFM using the same **650M**-parameter architecture. Evaluated additive profile embeddings, Perceiver cross-attention, and profile-based initialization for DFM.
- **Training and sampling:** Processed a protein MSA corpus at the tens-of-millions scale; trained with Hydra / Lightning, multi-GPU DDP, and bf16. Compared Langevin, ODE, and SDE sampling for BFN and analyzed the effects of step count and starting time.
- **Evaluation:** On CAMEO-61, mean pLDDT improved from **56.46** to **78.24** and mean pTM from **0.55** to **0.75** versus the previous-generation model. Used CCMpred long-range P@L5 to assess coevolutionary contact quality. These are **structure-prediction proxy metrics**.
- **Property-controlled editing (Tencent):** Contributed to conditional modeling and training, sampling, and evaluation pipelines; compared editing methods by target condition satisfaction, extent of sequence edits, and predicted structural quality.

Selected Research Projects (continued)

3D Molecular Conformation Generation: Geometric Diffusion and Efficient Sampling TPAMI

- **Distance distribution modeling (SDDiff):** For 3D conformation generation from molecular graphs, characterized the transition of perturbed interatomic distance distributions from Gaussian to Maxwell–Boltzmann. Proposed a shifting-score diffusion model accounting for rotational / translational symmetry and geometric feasibility.
- **Geometric sampling (TPAMI):** Used differential geometry to analyze the relationship between coordinate and interatomic-distance manifolds; derived projection-free reverse SDEs / ODEs and compared conformation coverage and matching accuracy across sampling step counts.
- **Evaluation:** Evaluated conformation coverage (COV) and matching error (MAT) on GEOM-QM9 / GEOM-Drugs. The SDDiff preprint reported mean **COV of 90.68% and MAT of 0.8564 Å** on GEOM-Drugs, with a COV threshold of 1.25 Å.

Generating Physical Dynamics under Priors ICLR 2025

- **Approach:** Incorporated distributional priors such as rotational / translational invariance, together with physical constraints such as conservation laws and PDEs, into diffusion generation to reduce inconsistencies between generated trajectories and physical laws.
- **Implementation and validation:** Handled linear and nonlinear constraints through constraint decomposition; validated the physical feasibility of generated trajectories and flow fields across multiple physical systems and PDE datasets.

Publications

* Equal contribution; + Project supervision; # Corresponding author. Name in bold.

On Diffusion Process in SE(3)-invariant Space.
Zihan Zhou, Ruiying Liu, Jiachen Zheng, Xiaoxue Wang, Tianshu Yu[#]. **IEEE TPAMI**.

Order Matters in Retrosynthesis: Structure-aware Generation via Reaction-Center-Guided Discrete Flow Matching.
Chenguang Wang[*], **Zihan Zhou**[*], Lei Bai, Tianshu Yu[#]. **ICML 2026**.

Incomplete Data, Complete Dynamics: A Diffusion Approach.
Zihan Zhou, Chenguang Wang, Hongyi Ye, Yongtao Guan, Tianshu Yu[#]. **ICLR 2026**.

TEMPO: Temporal Multi-scale Autoregressive Generation of Protein Conformational Ensembles.
Yaoyao Xu, Di Wang, **Zihan Zhou**, Tianshu Yu[#], Mingchen Chen[#]. **NeurIPS 2025**.

Generating Physical Dynamics under Priors.
Zihan Zhou, Xiaoxue Wang, Tianshu Yu[#]. **ICLR 2025**.

Learning to Decouple Complex Systems.
Zihan Zhou, Tianshu Yu[#]. **ICML 2023**.

Preprints

Learning Field Reconstruction from Incomplete Data by Globally Correcting Local Estimates.
Renhao Zhong, **Zihan Zhou**[+], Chiyuan Ma, Tianshu Yu. **arXiv preprint, 2026**.

Differentiable mass transfer modeling enables wet lab validated inverse analysis of liquid chromatography.
Baochen Li[*], **Zihan Zhou**[*], Yanqing Yu, Sen Lin, Peng Wang, Tianshu Yu[#], Xiaoxue Wang[#]. **ChemRxiv preprint, 2026**.

AMix-2: Establishing Protein as a Native Modality in Large Language Models.
Keyue Qiu[*], Yixin Wu[*], **Zihan Zhou**, Changze Lv, Lihao Wang, et al., Hao Zhou[#]. **arXiv preprint, 2026**.

Observation-Aligned Mask Priors for Learning Physical Dynamics from Authentic Occlusions.
Chiyuan Ma, **Zihan Zhou**[+], Tianshu Yu[#]. **arXiv preprint, 2026**.

Physics-Mediated Diffusion: Leveraging Intermediate Field Representations for Sparse PDE Inversion.
Zihan Zhou[*], Chiyuan Ma[*], Tianshu Yu[#]. **Preprint, 2026**.

Molecular Conformation Generation via Shifting Scores.
Zihan Zhou, Ruiying Liu, Chaolong Ying, Ruimao Zhang, Tianshu Yu[#]. **arXiv preprint, 2023**.

Technical Skills

- **Generative modeling:** Diffusion models, BFN, discrete flow matching, DiT, conditional generation, MSA-conditioned modeling, and sampling strategies.
- **Training and experimentation:** Lightning, Hydra, multi-GPU DDP, bf16; model scaling comparisons, ablation studies, and reproducible evaluation.
- **Scientific computing:** Physical priors and conservation constraints, forward / inverse PDE problems, SE(3) invariance, molecular conformation generation, and protein generation.